Problem Scenario: "When the Heart Can't Keep Up"

Portrait of Clara Lou

Clara Lou

Clara Lou, a 62-year-old woman, visits the clinic complaining of increasing fatigue, shortness of breath on mild exertion, and swelling of her feet over the past two weeks. She says she can no longer climb stairs without stopping to rest.

Medical History

Long-standing hypertension (15 years)

Previous myocardial infarction (5 years ago)

Smoker for 30 years

Physical Findings

Blood pressure: 165/90 mmHg

Pulse: 110 bpm

Respiratory rate: 24/min

Crackles at the lung bases

Bilateral ankle edema

Mild jugular venous distension

Investigations

Chest X-ray: Enlarged cardiac silhouette, pulmonary congestion

Echocardiogram: Left ventricular ejection fraction 35%

Discussion Prompts

What are the risk factors that contributed to the development of heart failure in Clara Lou.

Identify and describe the body’s compensatory mechanisms activated in heart failure. How do these mechanisms initially help but later worsen the condition?

Relate Clara Lou clinical manifestations (e.g., shortness of breath, edema, fatigue) to the underlying pathophysiological changes in the heart and circulatory system.

Based on her findings, which type of heart failure is most likely predominant in Clara Lou? Explain your reasoning using evidence from the case.